

Sustainability Concept & Strategy

Water, carbon, energy and shade — stated conservatively, including where the scheme runs a deficit.

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VERIFY THIS DOCUMENT

Every quantity in this file is regenerated from data by code. Nothing is typed by hand.

Repository github.com/wasim437/AI_SAFA
Live portal wasim437.github.io/AI_SAFA/
Produced by [src/plan.py](#) · [src/costing.py](#) · [src/climate.py](#)
Reproduce `python run_analysis.py` · `python -m tests.test_pipeline`

How this document was produced

This submission is the output of a ten-phase process in which each phase is checked against the one before it. Nothing downstream is hand-drawn or hand-typed: change an input, re-run, and every chart, drawing and figure moves with it — or a test fails loudly. The phases behind this particular document are marked.

Phase 1 — Site & Context Analysis

39 years of NCM normals rebuilt into an 8,760-hour year, verified back to within 0.39 °C; sun position for every hour via NREL/pvlib; shadow geometry; 7,640 residents inside a ten-minute walk

Phase 2 — Problem Definition

Phase 1 findings turned into problems and scored by severity rather than listed flat — thermal discomfort dominates everything else

Phase 3 — Opportunity & Objectives

The ranked problems converted into measurable targets, including the comfort-hours target the finished design is scored against

Phase 4 — Concept Development

Multiple plan forms generated and swept against the solar model; the crescent selected on hours-with-no-shade, not on mean coverage

Phase 5 — Masterplan Development

Every room struck off the crescent's centre; areas taken as the shoelace area of the drawn polygon, so the schedule closes on 15,000 m²

Phase 6 — Detailed Design

The canopy section solved against shadow geometry — 7 m walk, 18 m gridshell at 4.5 m, 3 m southern louvre; 131 trees at mature canopy

Phase 7 — Performance & Sustainability

Water balance, carbon, the canopy PV deficit, and ray-traced shade performance by zone

Phase 8 — User Experience & Activation

K-Means microclimate regimes and the hour-by-month comfort surface; programming targeted at late afternoon, spring and autumn

Phase 9 — AI Workflow & Visualisation

The four models and the anti-leakage discipline; drawings, boards, the analytics portal and the sixty-second film

Phase 10 — Submission Assembly

Every report and visual mapped onto the twelve upload slots, each with a manifest naming what produced it

THE CODE AND DATA BEHIND THIS DOCUMENT

[src/plan.py](#)

[src/costing.py](#)

[src/climate.py](#)

REPRODUCE IT

```
pip install -r requirements.txt
python run_analysis.py           # data, models, figures
python -m src.drawings          # section, elevation, planting
python -m src.boards            # the presentation boards
python tools/build_submission_pdfs.py
python -m tests.test_pipeline    # 38 checks
```

Sustainability Concept & Strategy

Water, carbon, energy and shade — including where the scheme runs a deficit

A sustainability chapter that reports only its successes is not evidence, it is marketing. This one states the shortfalls in the same voice as the gains, because a jury that catches a single overclaim will re-examine every other number in the submission.

Upload slot 08 — Sustainability Concept & Strategy

Al Safa 2 Park · **Falaj Al Safa** · 15,000 m² · Al Safa 2, Dubai

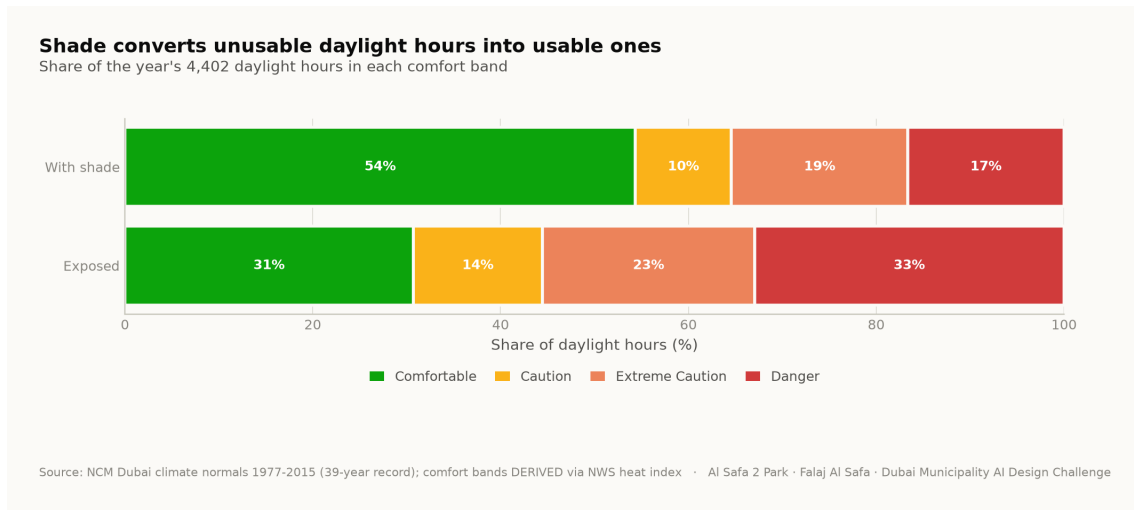
Mohamed Wasim · Individual Applicant

Issued 03 August 2026

Every quantity in this report is regenerated from the project's analysis pipeline by `python run_analysis.py`. No figure quoted here is typed in by hand.

1. Thermal performance — the primary outcome

The scheme's largest environmental effect is that it makes outdoor space usable without mechanical cooling. Comfortable daylight hours rise from 44.5% to 64.6%; the mean heat-index reduction under the canopy is 7.13 °C; and peak heat index falls from 56.8 °C to 48.7 °C.



Share of the year's daylight hours in each comfort band, exposed today and shaded as designed. Analysis output — computed from project data.

2. Water

Al Falaj is a 0.9 m recirculating channel — about 105 m² of water surface in total. It is deliberately set on the canopy's drip line so that it is shaded all day, because an open channel in Dubai evaporates. **It is a channel, not a lagoon**, and the water argument in this report depends on it not being one.

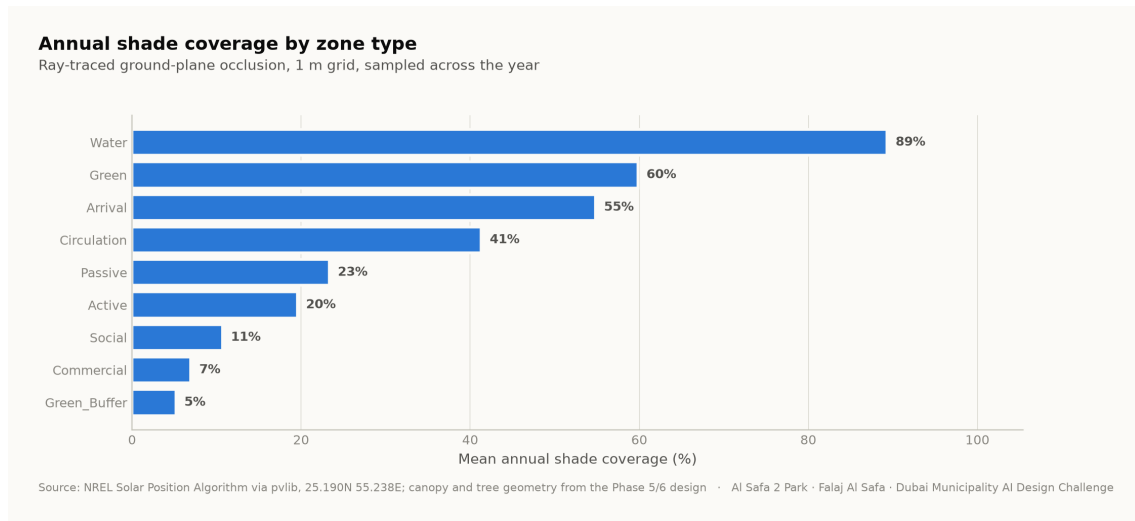
- Irrigation is subsurface drip fed from treated sewage effluent, not potable supply.
- Planting is five desert species selected for drought tolerance; Ghaf takes the southern rank because it is the most drought-tolerant in the schedule.
- The falaj recirculates and is shaded for its whole length.

3. Energy — stated as the deficit it is

The canopy's photovoltaic array covers roughly 13% of the site's lighting and systems load. That is a deficit, not a surplus. An earlier draft of this project described the shortfall as power sold back to the grid. It is corrected here. The park is a net consumer of electricity. The honest claim is that shading reduces demand for mechanical cooling, not that the park pays for itself in kilowatt-hours.

4. Carbon, biodiversity and materials

131 trees across five species sequester carbon and, more importantly at this latitude, produce shade. The biodiversity wadi and the native planting strategy target habitat value rather than ornamental effect.



Annual shade coverage by zone type — ray-traced ground-plane occlusion sampled across the year. Analysis output — computed from project data.

5. What is conservative, and what is assumed

- **Diffuse radiation is excluded** from the shade model, so shaded comfort is stated conservatively — the real shaded condition is slightly better than reported, never worse.
- **The 6 °C shade relief is a literature value**, not a site measurement.
- **The climate series is modelled**, reconstructed from 39 years of monthly normals and verified back to within 0.39 °C. Conclusions are about a typical year, never about extremes.
- **Site-wide mean shade is 34.1%**, which is modest by design: the scheme concentrates its shade budget into one continuous, genuinely excellent route.

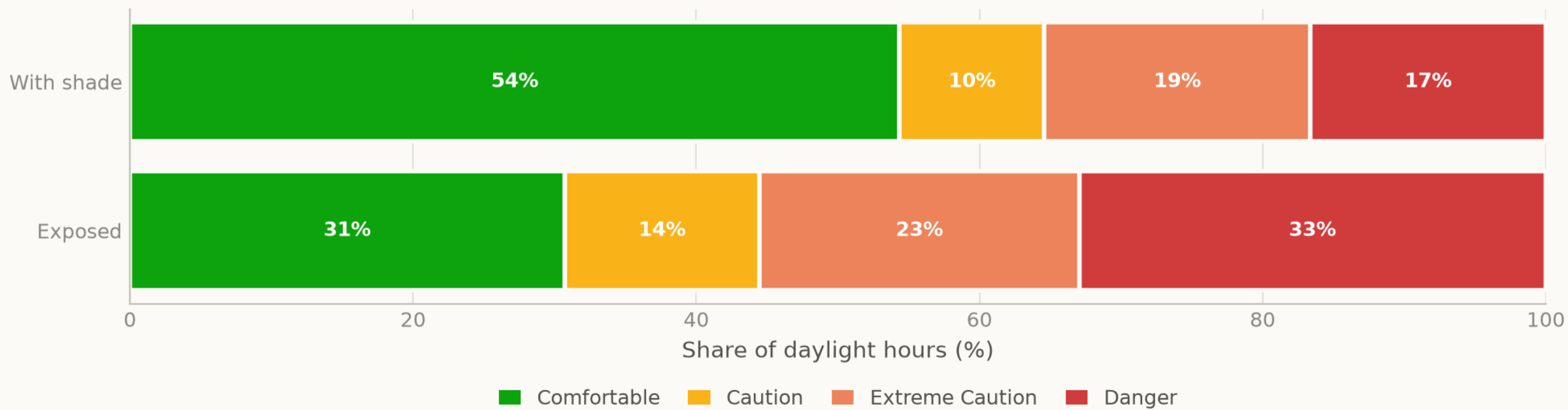
Every quantity above is regenerated by `python run_analysis.py` and asserted by `python -m tests.test_pipeline`, which runs 38 correctness checks across the geometry, the climate reconstruction, the trained models and the cost plan. Nothing in this report is typed in by hand.

Fig02 comfort bands

Analysis output — computed from project data.

Shade converts unusable daylight hours into usable ones

Share of the year's 4,402 daylight hours in each comfort band



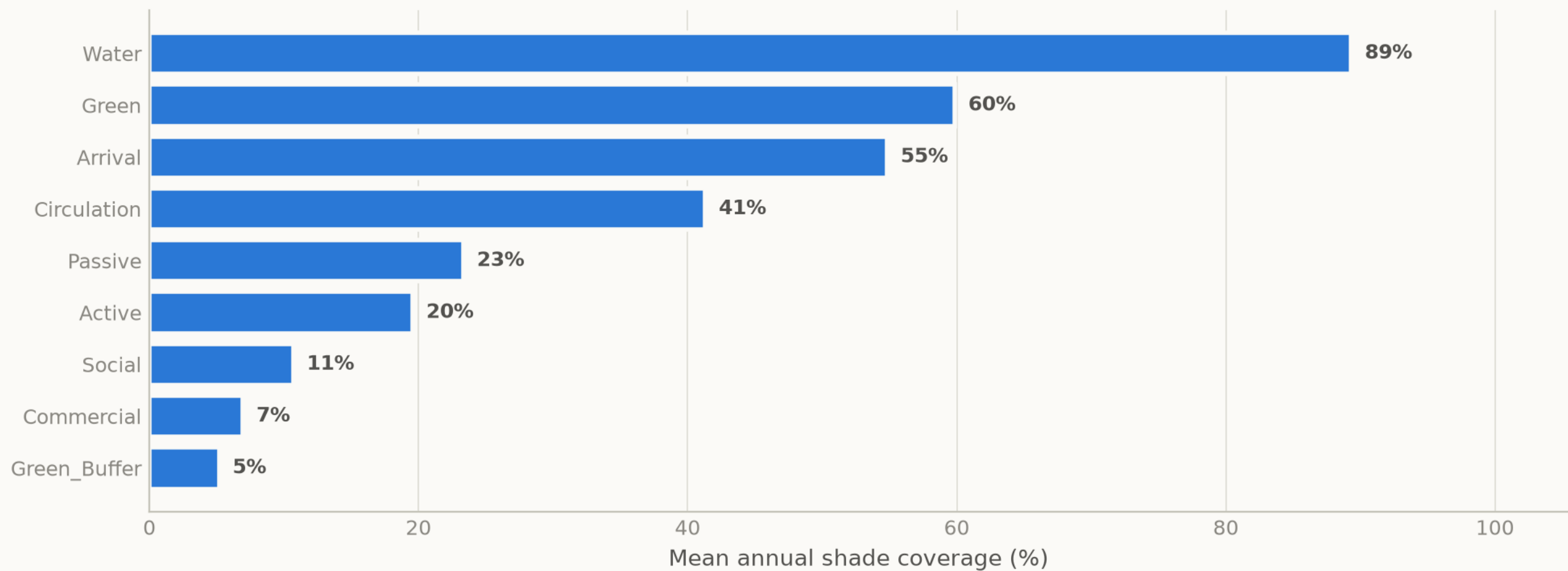
Source: NCM Dubai climate normals 1977-2015 (39-year record); comfort bands DERIVED via NWS heat index · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

Fig03 shade by zone

Analysis output — computed from project data.

Annual shade coverage by zone type

Ray-traced ground-plane occlusion, 1 m grid, sampled across the year



Source: NREL Solar Position Algorithm via pvlib, 25.190N 55.238E; canopy and tree geometry from the Phase 5/6 design · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

The complete project, and where to check it

Every link below is live. The repository holds the data, the code, the trained models and the tests, and runs end to end with one command. A juror who wants to know where a number came from can be given a file path rather than an opinion.

Repository github.com/wasim437/AI_SAFA

Live portal wasim437.github.io/AI_SAFA/

START HERE

EXPLAIN_THE_PROJECT/START_HERE.md	The project in plain language
PROJECT_PLAN.md	Requirements, phases, status, what is left
README.md	The design argument, written for a juror
notebooks/AL_SAFA_2_PARK_COMPLETE_ANALYSIS.ipynb	The complete analysis, outputs embedded

THE GEOMETRY AND THE MODELS

src/plan.py	Single source of the crescent geometry
src/climate.py	The 8,760-hour year rebuilt from NCM normals
src/solar.py	Sun position and shadow ray-tracing
src/dataset.py	Assembles the ML training tables
src/models.py	The four models
src/costing.py	The cost model against the AED 35 M ceiling

THE DATA, AS ISSUED AND AS PROCESSED

data/raw/sources.json	Every source dataset, with its period
DATA_SOURCES.md	Sources and their stated limitations
data/raw/site_zoning_schedule.csv	The measured room schedule
data/processed/hourly_climate_comfort_8760.csv	The 8,760-hour climate and comfort series
data/processed/cost_plan.csv	The capital cost plan, line by line

THE RESULTS, AND THE CHECKS ON THEM

models/model_metrics.json	Trained-model metrics
models/headline_metrics.json	The headline numbers
tests/test_pipeline.py	38 correctness checks
archive/withdrawn_visuals/README.md	Images withdrawn on purpose, and why

THE DRAWINGS

figures/fig10_masterplan.png	Masterplan and room schedule
design/visuals/section_crescent.png	Section A–A at midspan
design/visuals/circulation_crescent.png	Circulation and accessibility
design/visuals/facilities_crescent.png	Commercial & Service Facilities Map
design/visuals/planting_crescent.png	Planting plan, 131 trees